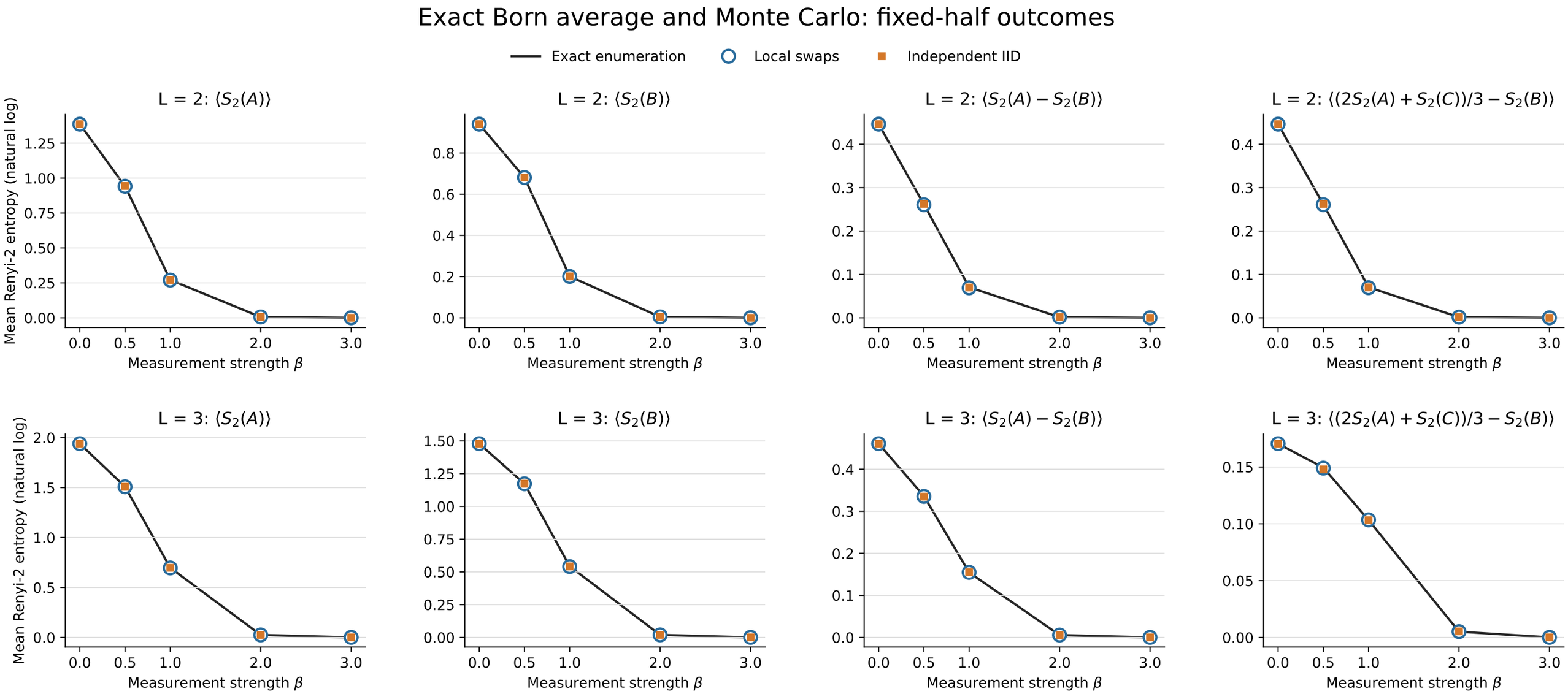
Exact Born average and Monte Carlo: fixed-half outcomes
Exact enumeration
Local swaps
Independent IID
L = 2: ⟨$S_2(A)$⟩
L = 2: ⟨$S_2(B)$⟩
L = 2: ⟨$S_2(A) - S_2(B)$⟩
L = 2: ⟨$(2S_2(A) + S_2(C))/3 - S_2(B)$⟩
L = 3: ⟨$S_2(A)$⟩
L = 3: ⟨$S_2(B)$⟩
L = 3: ⟨$S_2(A) - S_2(B)$⟩
L = 3: ⟨$(2S_2(A) + S_2(C))/3 - S_2(B)$⟩
Mean Renyi-2 entropy (natural log)
Measurement strength β
30,000 samples per method and case; bars = 2 empirical block-SE. Exact lines connect the calculated beta values.
Overlapping means do not establish reliable uncertainty: the L2, beta=3 IID run misses rare high-entropy outcomes (see residual figure).
L2 uses test regions (A: 1x2, B: 1x1, C: 2x1); L3 uses A: 1x3, B: 1x2, C: 3x1. No scaling coefficient is fitted.